ALEXANDRA BREMERS

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I study how machines can collaborate with people in the physical world. My approach combines design activities with systems prototyping and user studies. Prior to joining Cornell, I studied and developed future AI and AR human-machine interfaces at Jaguar Land Rover.

EDUCATION

2020- **Cornell University**, Cornell Tech, New York, NY, USA
Ph.D. in Information Science
Specialization: Human-Computer Interaction
Minor: Computer Science
Committee: Wendy Ju (chair), François Guimbretière, Stephen Marschner
Expected graduation date: August 2025

2016-2018 **Utrecht University**, Utrecht, The Netherlands
M.Sc. in Artificial Intelligence
Track: Cognitive Processing
Thesis: *Perception of Perspective in Augmented Reality Head-Up Displays*
Advisors: Chris Janssen, Ignace Hooge, Valerian Meijering

2013-2016 **Eindhoven University of Technology**, Eindhoven, The Netherlands
B.Sc. in Industrial Design
Graduation Project: *How to embroider a radio*
Advisors: Oscar Tomico, Saskia Bakker

2007-2013 **Stedelijk Gymnasium Arnhem**, Arnhem, The Netherlands
Gymnasium (Grammar School)

INDUSTRY RESEARCH EXPERIENCE

2021 (summer) **Research Intern**, Toyota Research Institute
Los Altos, CA, USA

2017–2020 **Human-Machine Interface Researcher**, Jaguar Land Rover
Warwick University, Coventry, United Kingdom

ACADEMIC RESEARCH EXPERIENCE

2020– **Graduate Research Assistant**, Future Automation Research Lab
Cornell Tech, Cornell University, New York, NY, USA
Collaborations with Amada AI Innovation Laboratory Ltd., Accenture Labs, Toyota Research Institute and Nissan Motors.

2022 (summer) **Research Visitor**, User Interfaces Group, Department of Engineering
Aalto University, Espoo, Finland

2021 (fall) **Research Visitor**, Robots in Groups Lab
Cornell University, Ithaca, NY, USA

2017 (summer) **Research Intern**, NTU IoX Center
National Taiwan University, Taipei, Taiwan

PUBLICATIONS

Journal Articles

Alexandra W.D. Bremers, Ali Özgür Yöntem, Kun Li, Daping Chu, Valerian Meijering, Christian P. Janssen. "Perception of Perspective in Augmented Reality Head-Up Displays". In: *International Journal of Human-Computer Studies)* (2021), p. 102693. ISSN: 1071-5819. DOI: https://doi.org/10.1016/j.ijhcs.2021.102693. URL: https://www.sciencedirect.com/science/article/pii/S1071581921001117.

Conference Papers

Avital Dell'Ariccia, **Alexandra Bremers**, Johan Michalove, Wendy Ju. "How to make people think you're thinking if you're a drawing robot". In: *Proceedings of the 17th ACM Conference on Human-Robot Interaction - Videos*. 2022. DOI: https://doi.org/10.1109/HRI53351.2022.9889638. URL: https://ieeexplore.ieee.org/abstract/document/9889638.

David Goedicke, **Alexandra Bremers**, Sam Lee, Fanjun Bu, Hiroshi Yasuda, Wendy Ju. "XR-OOM: MiXed Reality Driving Simulation With Real Cars". In: *Proceedings of the ACM International Conference on Human Computer Interaction (CHI)*. 2022. DOI: https://doi.org/10.1145/3491102.3517704. URL: https://dl.acm.org/doi/abs/10.1145/3491102.3517704.

Extended Abstracts

Alexandra Bremers, Maria Teresa Parreira, Wendy Ju. "Understanding Bystander Facial Responses to Robot Task Failures with the BAD Dataset". In: *HRI Workshop Position: The Imperfectly Relatable Robot*. 2023.

Alexandra Bremers. "A computer that sketches along with you". In: *Creativity and Cognition*. C&C '22. Venice, Italy, 2022. DOI: https://doi.org/10.1145/3527927.3533732. URL: https://dl.acm.org/doi/abs/10.1145/3527927.3533732.

Alexandra Bremers. "How can a robot help people draw?" In: *Companion Publication of the 2021 ACM Designing Interactive Systems Conference*. DIS '22 Companion. Virtual Event: Association for Computing Machinery, 2022. DOI: https://doi.org/10.1145/3532107.3532876. URL: https://dl.acm.org/doi/abs/10.1145/3532107.3532876.

Avital Dell'Ariccia, **Alexandra Bremers**, Wen-Ying Lee, Wendy Ju. "Work in Progress: "Ah! He wants to win!": Social responses to playing Tic-Tac-Toe against a physical drawing robot". In: *Adjunct Proceedings of the 2022 ACM International Conference on Tangible and Embodied Interaction*. 2022. DOI: https://doi.org/10.1145/3490149.3505571. URL: https://dl.acm.org/doi/abs/10.1145/3490149.3505571.

Natalie Friedman, Kari Love, **Alexandra Bremers**, AJ Parry, Ray LC, Bolor Amgalan, Jen Liu, Wendy Ju. "Workshop: Designing Functional Clothing for Human-robot Interaction". In: *Companion of the 2021 ACM/IEEE International Conference on Human-Robot Interaction*. 2021, pp. 703–705. DOI: https://doi.org/10.1145/3434074.3444870. URL: https://dl.acm.org/doi/abs/10.1145/3434074.3444870.

David Goedicke, **Alexandra Bremers**, Hiroshi Yasuda, Wendy Ju. "Work in Progress: XR-OOM: Mixing Driving Environments with Virtual Objects Safely". In: *Adjunct Proceedings of the 2021 ACM International Conference on Automotive Interfaces and Vehicular Applications*. 2021. DOI: https://doi.org/10.1145/3473682.3480266. URL: https://dl.acm.org/doi/abs/10.1145/3473682.3480266.

Under Review

Alexandra Bremers, Alexandria Pabst, Maria Teresa Parreira, Wendy Ju. "Using Social Cues to Recognize Task Failures for HRI: A Review of Current Research and Future Directions". In: *arXiv:2301.11972*. 2023. DOI: https://doi.org/10.48550/arXiv.2301.11972.

Alexandra Bremers, Maria Teresa Parreira, Xy Fang, Natalie Friedman, Adolfo Ramirez-Aristizabal, Alexandria Pabst, Mirjana Spasojevic, Mike Kuniavsky, Wendy Ju. "The Bystander Affect Detection (BAD) dataset for failure detection in HRI". In: *arXiv:2303.04835*. 2023. DOI: https://doi.org/10.48550/arXiv.2303.04835.

Nermin Caber, Jiaming Liang, Bashar I. Ahmad, Simon Godsill, **Alexandra Bremers**, Philip Thomas, David Oxtoby, Lee Skrypchuk. "Driver Profiling and Bayesian Workload Estimation for Adaptive In-Vehicle HMI". In: *IEEE Transactions on Intelligent Transportation Systems*. 2023.

Natalie Friedman, Kari Love, Awsaf Ahmed, **Alexandra Bremers**, Wendy Ju. "Designing a utility belt for a farm robot". In: *ACM Designing Interactive Systems Conference*. 2023.

David Goedicke, Natalie Chyi, **Alexandra Bremers**, Stacey Li, James Grimmelmann, Wendy Ju. "Mutual Benefit: The Case for Sharing Autonomous Vehicle Data with the Public". In: *Proceedings of the ACM International Conference on Human Computer Interaction (CHI)*. 2023.

Working Papers

Alexandra Bremers, Fanjun Bu, David Goedicke, Lunshi Zhou, Wendy Ju. "Detecting Confusion from Body Pose Data". In: *TBD*. 2023.

Alexandra Bremers, Natalie Friedman, Sam Lee, Tong Wu, Eric Laurier, Malte Jung, Jorge Ortiz, Wendy Ju. "Examining Discomfort in Shared Car-Rides". In: *TBD*. 2023.

Alexandra Bremers, Francois Guimbretiere, Wendy Ju. "ClayWOz – Platform for Remote and In-Person Collaborative Ceramics". In: *TBD*. 2023.

Alexandra Bremers, Wendy Ju, Hiroshi Yasuda. "People on the Street: Sampling Real-Life through Novel Large-Scale Datasets". In: *TBD*. 2023.

Alexandra Bremers, Ruidong Zhang, Xy Fang, Fanjun Bu, Natalie Friedman, Adolfo Ramirez-Aristizabal, Alexandria Pabst, Mirjana Spasojevic, Mike Kuniavsky, Cheng Zhang, Wendy Ju. "Detecting Robot Task Failures Through Bystander Facial Expressions Using Neck-Mounted Wearables". In: *TBD*. 2023.

Cooper Murr, **Alexandra Bremers**, Wendy Ju. "Putting it to paper: exploring creative algorithms with ShadeBot". In: *TBD*. 2023.

Tobias Weinberg, **Alexandra Bremers**, Wendy Ju. "AxiWOz – Experimental Platform for Collaborative Drawing Studies with the AxiDraw Pen Plotter". In: *TBD*. 2023.

Lunshi Zhou, **Alexandra Bremers**, Wendy Ju. "Work in Progress: Initiative and Materiality: Exploring Mixed Reality Calculators with the Tangible Human-A.I. Interaction Framework". In: *TBD*. 2023.

GRANTS, HONOURS & AWARDS

Fellowships

Google Fellowship Departmental Nominee
Date: 2022
Cornell University Bowers CIS - Information Science.

Royal Commission of 1851 Industrial Fellowship
Amount: £90,000
Date: 2020
Fellowship to pursue a PhD at the University of Cambridge Department of Engineering. Declined in order to accept Cornell offer.

Grants

Finnish Center for AI (HIIT/FCAI) Travel Support Grant
Amount: €3,500

Date: 2022

ACM Creativity & Cognition Graduate Symposium NSF Student Travel Grant
Amount: $1,900
Date: 2022

Cornell Graduate School Conference Grant
Amount: $700
Date: 2022

Cornell Graduate School Conference Grant
Amount: $675
Date: 2021

Cornell Tech Backslash Art Microgrant: **Plotting with Conductive Inks.**
Amount: $500
Date: 2021

Honours & Awards

CRA-WP Grad Cohort for Women
Date: 2023

Cornell Tech Conversations: Women Who Lead (WWL) Inaugural Cohort
Date: 2022

TATA Innovista Global Finalist & JLR Innovista Global Finalist
Date: 2019

Oxford-Cambridge Rising Women in Science and Engineering
Date: 2018

Bruynzeel Storage Systems, Designing the Office Competition, 2nd Student Prize
Value: €500
Date: 2015

Schoolkrantawards, Best School Paper of The Netherlands (Content & Design)
Value: €500
Date: 2012

Lions Club Arnhem Arenacum & Stedelijk Gymnasium Arnhem, Best Report
Value: €200
Date: 2012

TALKS

Invited talks

Invited Speaker, "Drawing with Physical AI". Design Lab, University of Twente. 2023. (Upcoming)

Invited Speaker, "Mixed-Initiative HRI in a drawing task". Finnish Center for AI. 2023. (Upcoming)

Invited Speaker, "Interaction Intelligence for HRI". Accenture Labs California (online). With Wendy Ju. October 21st, 2022.

Invited Speaker, Cornell Information Science Seminar. April 2022.

Invited participant and panelist, "How can models inform design?". Computational Models of Human-Automated Vehicle Interaction Seminar. Schloss Dagstuhl – Leibniz-Zentrum für Informatik, Germany. March 6-11, 2022.

Invited Speaker, "After RisingWISE". RisingWISE reunion event by University of Oxford and University of Cambridge. Woburn Safari Park Lodge, United Kingdom. September 25th, 2021.

Guest Speaker, "XR-OOM: miXed Reality On-rOad siMulator". XR-Monthly by the XR-Collective at Cornell Tech (online). With David Goedicke. February 19th, 2021.

Invited Participant, "RisingWISE". Event series by University of Oxford and University of Cambridge to connect women in STEM research. Held between Saïd Business School Oxford and Madingley Hall Cambridge, United Kingdom. November 2018 - January 2019.

Other Presenter. Cornell Tech outreach (online). July 2022.

Presenter. Computing and Information Science Research Night. Gates Hall, Cornell University, Ithaca, NY. November 2, 2021.

ADVISING

2021– **Internship Advisor**, Cornell Tech, Cornell University, New York, NY

2022 Cooper Murr: *Translating digital algorithms to paper*. (UG, Cornell)

- Working on LBW submission to CHI'23.

2022 Daniel Zhou: *Exploring HAI interaction: calculator*. (UG, Cornell)

- Working on WIP submission to TEI'23.

2022 Tobias Weinberg: *Drawing together*. (UG, Technion)

- Working on submission to CSCW'23.

2022 Mary Lee: *Bad Robots*. (High School)

2021 Avital Dell'Ariccia: *Robotic Tic-Tac-Toe*. (UG, Technion, Israel)

- Work-in-Progress paper accepted at TEI 2022.
- Video accepted at HRI 2022.
- Awarded Gary Marshden Travel Grant to HRI and TEI.

TEACHING

2021– **Teaching Assistant**, Cornell Tech, Cornell University, New York, NY, USA

2022 (F) CS5424 Developing and Designing Interactive Devices

2021 (F) CS5424 Developing and Designing Interactive Devices

2021 (S) CS5678 Topics in Mixed Reality

DESIGN & ART EXPERIENCE

2015 (fall) **Speculative Design Intern**
Next Nature Network, Amsterdam, The Netherlands

2013–2015 **Volunteer at Interactive Art Exhibitions**
Dutch Design Week / Fiber Festival / GLOW Festival, The Netherlands

EDUCATION (OTHER)

2019 **Warwick University**, Coventry, United Kingdom
M.Eng. Module (5 CATS Credits), Human-Technology Interaction for Automotive

2010 **Radboud University**, Nijmegen, The Netherlands
Non-degree module, Designing and Making Web-Based Software

SERVICE

Departmental Service

2022-23

- Task Force Design+Technology Center and Initiative, Cornell

2021-22

- Task Force Design+Technology Center and Initiative, Cornell
- President of Information Science Graduate Student Association
- PhD admissions reviewer and interviewer

2020-21

- Task Force Design+Technology Center and Initiative, Cornell
- PhD admissions reviewer and interviewer

Academic Service

Active service in:

ACM DIS (Designing Interactive Systems)

2021 Social Media Chair

ACM AutoUI (Automotive User Interfaces and Interactive Vehicular Applications)

2020 Moderator

2019 Visual Design Chair

Peer Reviewer for:

2023 ACM THRI

2022 ACM CHI

2021 ACM AutoUI

2021 ACM IMWUT

2021 ACM DIS

2021 IEEE IV

2020 ACM CHI

2019 ACM Auto-UI

ASSOCIATIONS

Association of Computer Machinery

Optical Society of America

PROGRAMMING LANGUAGES

Proficient: Python, R
Experienced: C++ (Arduino), C# (Unity), Java (Processing)